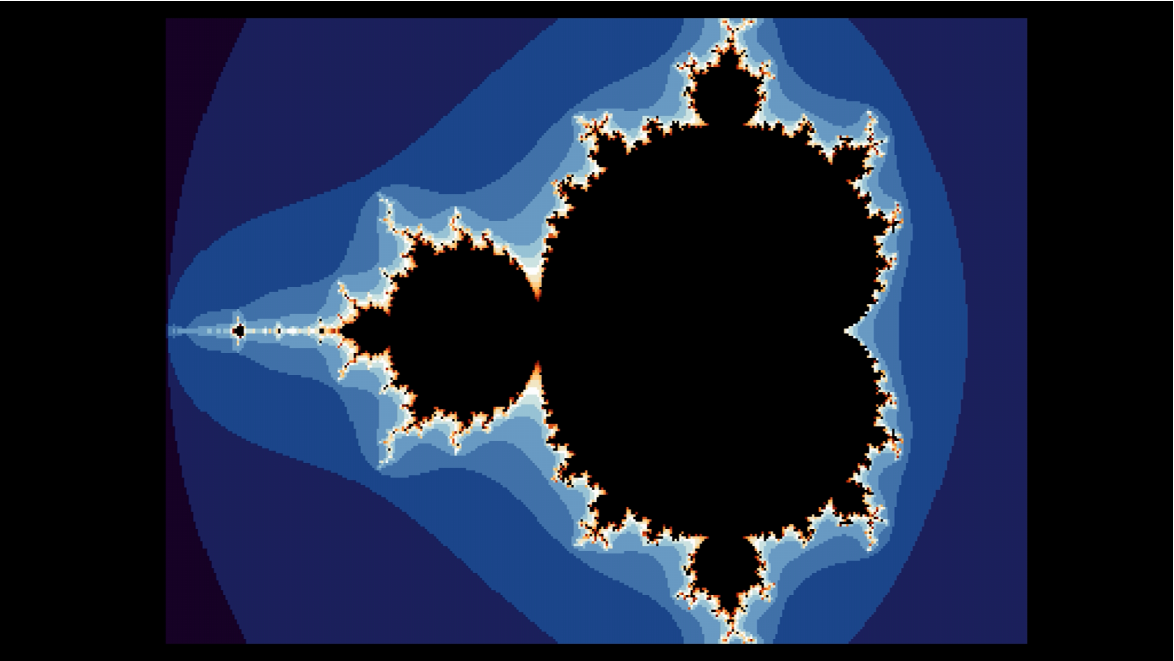


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Mandelbrot Upic



A Commodore 64 Ultimate demo that generates a Mandelbrot fractal on-device at 64 MHz turbo, packs it directly into Upic format (a 16-color, 384x256 border-color raster picture technique), displays it live as it renders, and lets you interactively pan around and zoom into any region of the result.

The same PRG also runs on an original Ultimate 64 or Ultimate 64 Elite (48 MHz turbo maximum). It measures the CPU speed at startup and, on a 48 MHz machine, shows 288 of the 384 pixels per line (3 of every 4, each about 1.36 dots wide) at the same size and screen position: every shown pixel is within 1.5 dots of where a 64 MHz machine shows it. The 48 MHz path has been checked on an Ultimate 64 Elite, against a C64 Ultimate running the same PRG (see `make e2e` under [Make targets](#)).

See `CREDITS.md` for full attribution.

Status: v1.0.3, feature-complete and confirmed working on real Ultimate 64 hardware (firmware 3.15 and 3.15a).

Watch it in action -- real-hardware capture: live generation, all 4 palettes, and interactive zoom.

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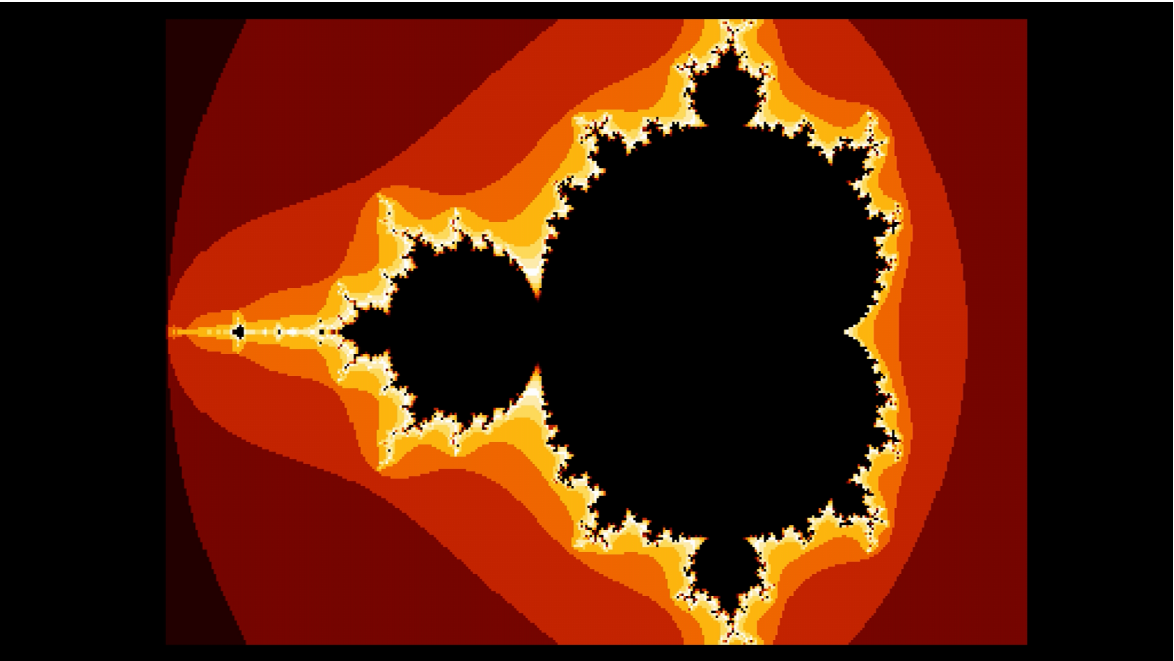
Controls

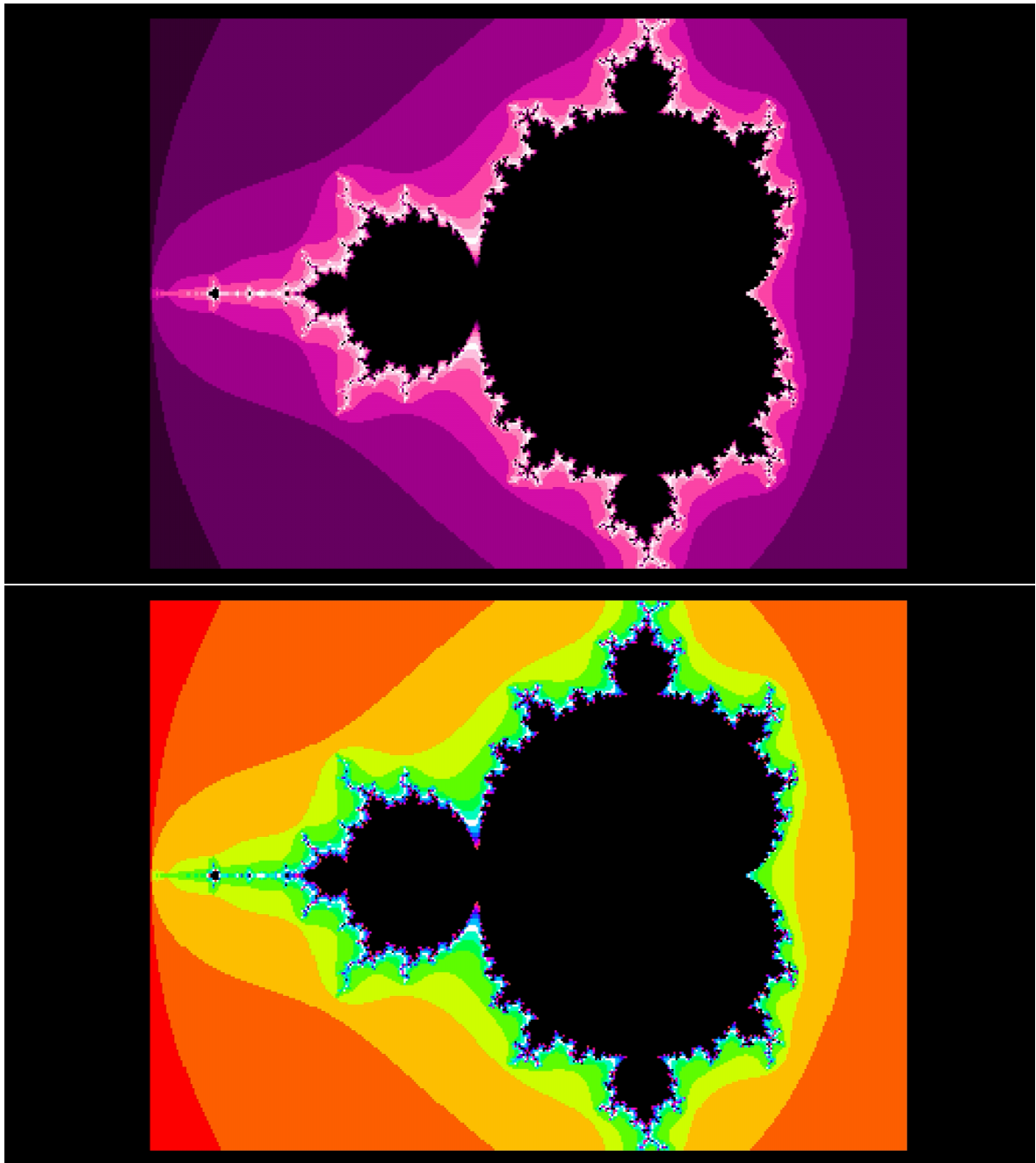
Generation starts automatically on launch and the picture builds up live, left to right. Once it completes, you're in **browse mode**:

Input	Action
W/A/S/D	Pan the current view (no-op at the default overview -- nothing to pan to)
Cursor keys (hold Shift for up/left)	Same, alternative for muscle memory
Z	Enter box mode to pick a zoom target
O	Zoom out one notch (widens the view, clamped to the original overview)
C	Cycle the base color gradient (sunset, fire, amethyst, rainbow)

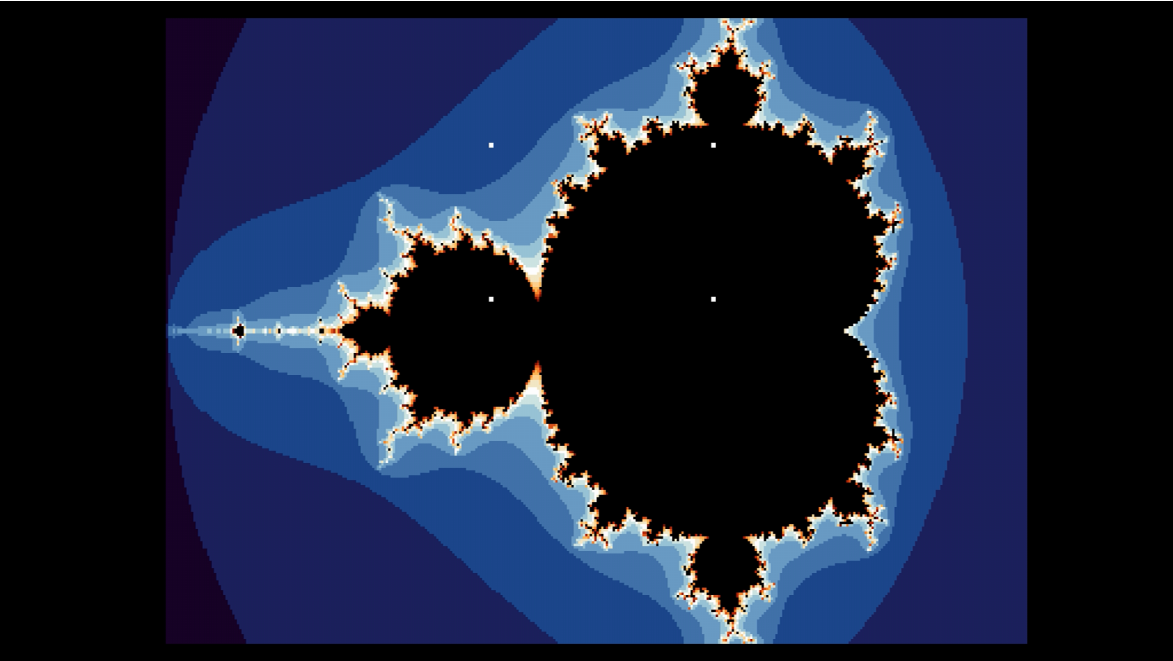
Panning regenerates the fractal at the same zoom level, shifted -- each step is a fraction of the current view's own size, so it moves less in absolute terms the deeper you've zoomed in.

The 4 selectable gradients:



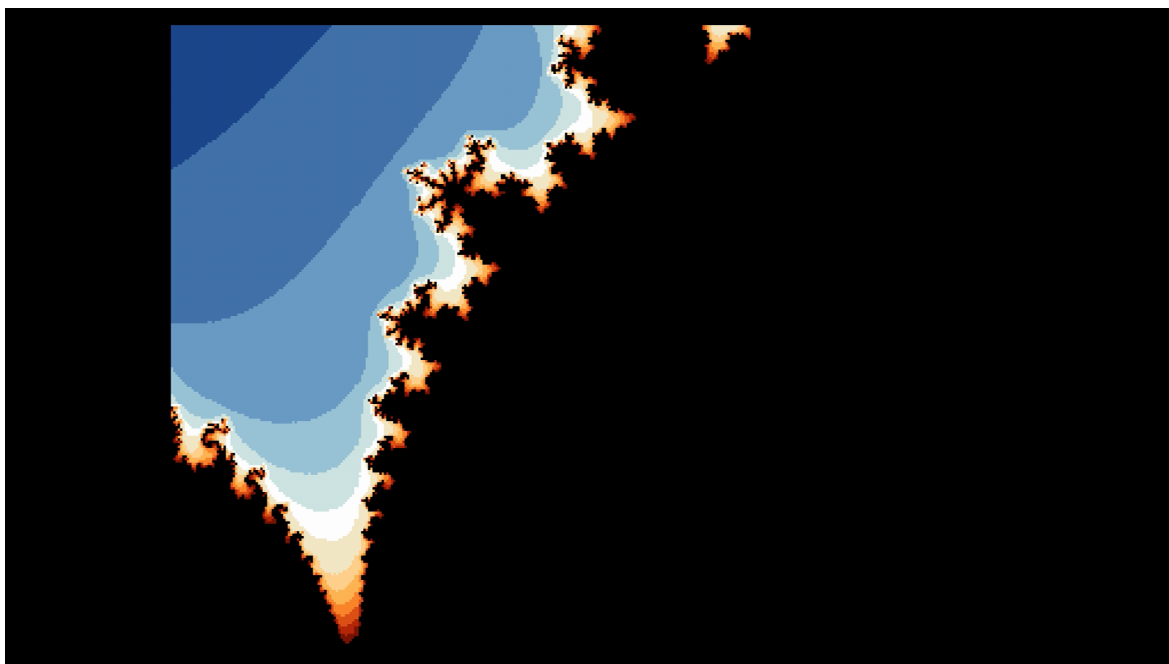


Press Z to enter **box mode**: 4 corner markers (solid white 2x2 blocks) appear, outlining a box that always keeps the picture's own 3:2 aspect ratio:



Input	Action
W/A/S/D	Move the whole box
Cursor keys (hold Shift for up/left)	Same, alternative for muscle memory
+	Grow the box (aspect ratio unchanged)
-	Shrink the box (aspect ratio unchanged)
RETURN	Confirm and zoom into the box
Z	Cancel back to browse mode without zooming
C / O	Same as browse mode

Confirming regenerates the fractal at the selected region and returns to browse mode. Repeated zooms compose relative to whatever's currently displayed, so zooming, panning, and zooming again all work together.



Zoom precision limit: the fractal coordinates use a fixed-point format with a finite number of fractional bits, capping how far repeated zooming can go (roughly 16x total from the initial view) before individual pixel steps round down to zero and the picture stops changing with further zoom. This is a hard limit of the current implementation, not a bug -- see docs/MANDELBROT_ALGORITHM.md.

No quit key -- reset or power off to exit, same as many C64 demos with no graceful exit path. See docs/ZOOM_FEATURE.md for why.

Installation

Requires **firmware 3.15 or newer**. As of this release, that means an **Ultimate 64 Elite 2** in practice -- the corresponding firmware for the original Ultimate 64 (C64U) board hasn't been released yet.

1. Copy both `mandelupic.prg` and `mandelupic.cfg` onto your Ultimate's SD card or USB storage, in the same folder -- extracting the release ZIP already places them together (at `idi8b/mandelupic/`), or make `deploy/make zip` produce them from source with matching names.
2. Run `mandelupic.prg` from the Ultimate's file browser.

The Ultimate's own firmware auto-loads a config file that shares its base name with the program being run -- `mandelupic.cfg` next to `mandelupic.prg` is picked up automatically, no manual "load config" step needed. It enables the Command Interface (UCI) and U64 turbo registers this demo needs (for an Ultimate 64 Elite 2 board); if your own configuration already has both enabled, this has no effect either way.

Known open point: the `.cfg` sets `Turbo Control` to `U64 Turbo Registers`. On a C64 Ultimate (firmware 1.2RC) that value is named `C64U Turbo Registers`, and the Ultimate's REST API rejects `U64 Turbo Registers` there. Whether the automatic `.cfg` load also skips the setting has not been checked. If the demo does not run at turbo speed on a C64 Ultimate, set `Turbo Control` to `C64U Turbo Registers` in the Ultimate menu.

Building from source

Prerequisites

Tool	Purpose	Install
Oscar64	C cross-compiler targeting 6502/C64	build from source, see their README
wput	FTP deploy to the Ultimate device	<code>sudo apt install wput</code>
pandoc + texlive-xetex	Generate README.pdf (optional)	<code>sudo apt install pandoc texlive-xetex</code>
Python 3	make test, make e2e (optional)	usually preinstalled

Deploy setup

Copy `.env.example` to `.env` and set `ULTIP1` to your Ultimate device's IP address (and, for make e2e, `E2E_DEVICES` to the devices to test on):

```
cp .env.example .env
```

Make targets

Target	Effect
<code>make / make all</code>	Compile to <code>build/mandelupic.prg</code> , regenerate <code>README.pdf</code> , build the release ZIP
<code>make deploy</code>	FTP the compiled <code>.prg</code> and matching <code>.cfg</code> to the Ultimate device set in <code>.env</code>
<code>make test</code>	Build both PRGs and run the host-side tests in <code>tests/</code> (Python 3, standard library only)
<code>make force48</code>	Build the test-only <code>build/mandelupic-force48.prg</code> , which always uses the 48 MHz display path (at 48 MHz), so that path can be checked on an Elite II / C64U
<code>make deploy-force48</code>	FTP the force48 PRG (as <code>mandelupic-force48.prg</code> , with its own copy of the <code>.cfg</code>) next to the release one
<code>make e2e</code>	Run the release PRG on the real devices listed in <code>E2E_DEVICES</code> in <code>.env</code> , drive it with key presses over the REST API, and compare the VIC video stream with the golden images in <code>tests/e2e/golden/</code> (see <code>tests/e2e/README.md</code>)

Target	Effect
<code>make e2e-update</code>	Same run, but write the captures as the new golden images for each device's display path
<code>make docs</code>	Regenerate README .pdf via pandoc
<code>make clean</code>	Remove build outputs

Documentation

Document	Covers
docs/ARCHITECTURE.md	Program flow, components, memory layout overview
docs/MANDELBROT_ALGORITHM.md	Fixed-point fractal generation algorithm
docs/UPIC_VIEWER.md	The Upic border-color raster display technique
docs/ZOOM_FEATURE.md	Interactive pan/zoom/palette control scheme
TURBOCONTROLMANUAL.md	Ultimate 64 CPU speed control library
UCILIBMANUAL.md	Ultimate Command Interface (UCI) protocol library
CHANGELOG.md	Version history
CREDITS.md	Full attribution